

Jurnal Kesehatan Gigi

p-ISSN: [2407-0866](#)e-ISSN: [2621-3664](#)<http://ejournal.poltekkes-smg.ac.id/ojs/index.php/jkg/index>

Development of A Clinical Decision Support System for Calculus Risk Prediction

Marlindayanti¹ Wahyu Jati Dyah Utami² Annisa Baizan³ Yasmin Tsaliza Muntazi⁴ Putri Sepriani⁵ Qori'Atun Nabilah⁶

^{1,4,5,6}Department of Dental Health, Poltekkes Kemenkes Palembang, Indonesia

²Department of Dental Health, Poltekkes Kemenkes Semarang, Indonesia

³School of Computer Science, Universitas Bina Nusantara, Indonesia

Corresponding author: marlindayanti@poltekkespalembang.ac.id

ABSTRACT

Oral health in Indonesia remains a significant problem, with 57.6% of the population experiencing problems but only 10.2% accessing treatment. Dental calculus is one of the main etiologies of periodontal disease. Therefore, healthcare professionals require a rapid and standardized screening tool to efficiently identify high-risk patients. This study aims to report the development process and trial results of the "Calrisk Dentistry" application, a clinical decision support system for detecting risk levels and identifying the main causative factors of calculus. This study used a cross-sectional design to conduct a trial of the system on 60 respondents. The trial was conducted on 60 patients, where the data collection process was facilitated by dental hygiene students. Data were collected using the "Calrisk Dentistry" application that analyzes 13 risk variables to generate a risk score. The application successfully classified respondents into four risk categories: Low (15%), Moderate (40%), High (31.7%), and Very High (13.3%). The risk factors with the highest impact identified by the system were inappropriate brushing time, one-sided chewing, and the presence of caries. The trial results indicate that the developed "Calrisk Dentistry" application is a functional and effective tool for calculus risk screening. This system has the potential to become a technological tool to increase patient awareness and facilitate early prevention.

Keyword : calculus; risk prediction; health application; decision support system; early prevention

Introduction

Oral health is widely recognised as an integral component of general health and significantly influences an individual's quality of life [1], [2]. Disorders of the oral cavity can impact overall health and an individual's social development. The 2023 Indonesia Health Survey showed that 56.9% of the Indonesian population experiences oral health, but only 11.2% access treatment [3]. The most common oral health problem is periodontal disease. One of the main etiological factors for periodontal disease is tartar (calculus), which is caused by public ignorance [4]. Periodontal disease can be prevented by regularly cleaning plaque and removing tartar. As many as

95.6% people in Indonesia brush their teeth every day, but only 6.2% brush their teeth in a proper time [3], indicating low public awareness and practice in maintaining oral health.

Calculus has emerged in recent literature as a major risk factor for periodontal inflammation and disease progression [5], [6]. The retention of subgingival calculus not only serves as a reservoir for biofilm but may itself trigger pathophysiologic pathways distinct from biofilm alone [7], [8]. This underscores the importance of early detection and removal of calculus in prevention efforts.

Several critical weaknesses in the conventional methods relies on the highly subjective, unstandardized, time-consuming and

inconsistent. Traditional approaches, which rely on clinical indicators such as probing depth, bleeding on probing, and plaque or calculus indices, remain highly subjective and prone to examiner bias [9]. Conventional models lack standardized development and are often misinterpreted as causal tools rather than predictive systems [10]. Moreover, conventional examinations remain time-consuming and inconsistent, especially across different clinical environments [11]. Additionally, barriers such as the high cost, lack of long-term validation, and limited clinician adoption further restrict the use of traditional risk assessment tools [12]. Hence, there is growing momentum towards digital technology to be involved [13], such as Clinical Decision Support Systems (CDSS) and risk-calculator frameworks to aid clinicians in timely, objective, and standardized risk assessment. These developments point to a broader technological shift: from purely clinical observation toward decision-support and automated risk stratification in dental care.

A study of using technological development to decide type 2 diabetes and periodontitis care shows that digital helps to improve classification and decision making in dental medical interface settings. It presents that under DSS assistance, General Practitioners had 8.3 times higher to recommending a visit to General Dental Practitioners. Although statistically, there is no significant impact of DSS assistance on the General Practitioners regarding common risk factors for type 2 diabetes and periodontal disease [14]. A mobile-phone diagnostic tool was also used to detect caries using YOLOv3 with Faster R-CNN which resulted in 87.4% and 71.4% sensitivity [15]. It shows that the average sensitivity is 79.4%. Separated from that, an AI-based detection using YOLOv8 of dental calculus has shown high precision in detecting hard-to-see calculus from bite-wing radiographs using deep learning. The system has a precision rate of 97.48% [16]. YOLOv8 has more enhancements than the previous versions of YOLOv5 or YOLOv7. However, detecting very small objects using that suggestion is not optimal, as the spatial detail of the object disappears [17], such as detecting plaque or calculus on small teeth in the intraoral photos.

To fill this gap and provide healthcare professionals with effective tools, this study reports the development and trial of "Calrisk Dentistry," a clinical decision support system implemented as an interactive web application. This system is designed as a rule-based expert system, where expert

knowledge and clinical logic are translated into a set of computational rules. Specifically, a quantitative scoring algorithm was developed to process 13 input variables—including clinical and behavioral data to generate personalized calculus risk predictions.

This study aims to describe the development process of the "Calrisk Dentistry" clinical decision support system and report the results of a trial of the system's effectiveness in classifying calculus risk levels and identifying dominant factors in a patient population.

Methods

This study used a cross-sectional study design with a descriptive-quantitative approach. This design was chosen to test the system's functionality and describe the level of calculus risk and dominant factors in a patient population at a single point in time. The study was conducted between September and November 2024.

The study participants were 60 patients selected using the convenience sampling method. The inclusion criteria for this study consisted of: (1) patients attending the clinic who were 18 years of age or older, (2) those who agreed to participate voluntarily and signed the informed consent form, and (3) individuals who were able to communicate effectively during the data collection process. Meanwhile, the exclusion criteria included patients who were unable to provide the complete data required for the questionnaire.

The main instrument used was the "Calrisk Dentistry" application, a clinical decision support system developed for this study. This study has been approved for its Ethical Clearance No. 1038/KEPK/Adm2/XI/2024.

System Architecture

This application is built as a web-based system using the PHP programming language for back-end logic. The user interface (front-end) is developed using HTML, CSS, and JavaScript. The application functions as a real-time calculator to generate printable analysis results.

Input Variables

The system processes 13 input variables consisting of: Calculus Index (CI), Saliva pH, Saliva Secretion, Caries, Tooth Arrangement, Toothbrushing Frequency, Toothbrushing Skills, Toothbrushing Time, Chewing Method, Smoking Habits, Number

of Cigarettes, History of Systemic Diseases, and Chronic Medication Consumption.

Analysis Mechanism and Scoring Algorithm

A specially designed quantitative scoring algorithm processes user input data. The risk calculation process is carried out in four stages. In the first stage, there was the Calculation of Individual Risk Scores. Each of the 13 input variables is converted into an individual risk score (scale 0-100) using the Binary Calculation for the variables Toothbrushing Time, Toothbrushing Skills, Chewing Technique, Systemic Diseases, and Chronic Medication Consumption. The ideal answer (input value 0) is given a risk score of 0. In contrast, the risky answer (input value 1) is given a risk score of 100. Next is a Proportional Calculation for the variables CI (maximum value 3), Caries (maximum value 3), Toothbrushing Frequency (maximum value 2), Smoking Habit (maximum value 3), Cigarettes (maximum value 3), and Tooth Arrangement (maximum value 4). The scores were calculated using Equation (1).

$$Score = \frac{\text{Input value}}{\text{Maximum value}} \times 100 \quad (1)$$

Next, there was a Reverse Calculation for the Saliva Secretion variable. The risk score is inversely proportional to the input value, calculated using Equation (2).

$$Score = \frac{(1.1 - \text{Input value})}{1.1} \times 100 \quad (2)$$

Subsequently, there was a Complex Calculation for the Salivary pH variable. The score is calculated based on the deviation from the normal range (6.8-7.2). If the pH is acidic (<6.8) or alkaline (>7.2), the risk score will increase significantly.

The second stage involved aggregating all individual risk scores. The total score was calculated by summing all thirteen risk values and dividing the total by thirteen, resulting in a final mean risk score expressed as a percentage representing the patient's overall calculus risk. The third stage consisted of risk classification, in which the mean percentage score was used to categorize patients into four distinct risk levels: low ($\leq 25\%$), moderate ($> 25\% \leq 50\%$), high ($> 50\% \leq 75\%$), and very high ($> 75\%$) [18]. Finally, in the fourth stage, the system identified the variable with the highest individual risk score to determine the most dominant contributing factor to the patient's overall calculus risk. Based on the results of the risk

classification and identification of dominant factors, the system automatically generates personalized preventive recommendations. These recommendations were derived from a knowledge base programmed into the application, where each risk factor has specific trigger conditions and predetermined messages.

User Interface

The application was designed with a simple workflow for healthcare professionals. The interface consists of three main sections, illustrated in the following figures.

Figure 1. Identity Page

Figure 1 shows the identity page which includes the name, age, gender and address of the patient.

Figure 2. Examination Page

Figure 2 shows the examination page, which is designed to collect clinical and behavioral data related to oral health. The form contains several input fields and dropdown menus corresponding to thirteen assessment variables used for calculating calculus risk. These include: Calculus Index (CI), saliva pH, saliva secretion, caries status, tooth alignment, brushing frequency, brushing skill, brushing time, chewing pattern, smoking frequency, number of cigarettes per day, systemic diseases, and long-term medication use. This structured input form enables users, particularly dental health professionals, to input patient data consistently and systematically, ensuring that the *Calrisk Dentistry* system can accurately generate individualized calculus risk scores based on the entered information.



Figure 3. Results Page of Risk Analysis

Figure 3 shows the Results Page, which automatically displays the risk analysis in the form

of diagrams, detailed score tables, and printable conclusions and recommendations.

Data Collection Procedure

After obtaining informed consent from the patient, the data collection process for each patient was facilitated by dental health students acting as enumerators. The students conducted interviews and basic examinations with the patients to obtain the required data for the 13 risk variables. Next, the students (as system users) input this data into the "Calrisk Dentistry" application. After the application generated the risk analysis results page, the results were then printed or manually recorded by the students. The collection of all these results constituted the research dataset.

Data Analysis

The research data recorded from the application output was analyzed using descriptive statistics. This analysis aimed to calculate the frequency distribution and percentage of each risk variable as well as the proportion of participants in each final risk category (Low, Medium, High, and Very High). The entire data analysis process was conducted using Microsoft Excel software.

Results and Discussion

This section presents the findings from the system trial of the *Calrisk Dentistry* application, which was developed as a digital clinical decision support tool for assessing calculus (tartar) risk in patients. The results describe the demographic characteristics of participants, the distribution of calculus risk levels generated by the system, and the dominant factors contributing to calculus formation identified through the application's analytical process.

Table 1. Respondents Characteristics

Characteristic	Category	n	%
Gender	Male	24	40
	Female	36	60
Age	Adolescents (17-25)	15	25
	Young adult (26-35)	25	41.7
	Adult (36-45)	12	20
	Elderly (>45)	8	13.3
	Total	60	100

According to Table 1, the system trial involved 60 patients, with the majority being female (60%), while male participants accounted for 40%. Most participants belonged to the young adult age group (26–35 years), representing 41.7% of the total sample. This demographic profile reflects a population that is generally active, health-conscious, and frequently visits dental clinics for preventive care. The distribution of participants' age and gender indicates a balanced sample that allows meaningful interpretation of behavioral and clinical risk factors related to calculus formation.

Table 2. Risk Category Distribution based on Application Detection

Risk Category	n	%
Low	9	15
Moderate	24	40
High	19	31.7
Very high	8	13.3
Total	60	100

Based on Table 2, the analysis using the "Calrisk Dentistry" application, 60 participants were distributed into four risk categories for calculus formation. The majority (40%) were in the moderate risk category.

Table 3. Influencing Factors Percentage of Calculus Risk

Risk Factor	Influencing Percentage (%)
Improper Toothbrushing Time	21
One-Sided Chewing Habit	18
Presence of Dental Caries	16
Presence of Calculus (CI)	12
Irregular Tooth Alignment	10
Alkaline Saliva pH	9
Toothbrushing Skill Level	8
Smoking Habit	4
Systemic Disease	2
Total	100

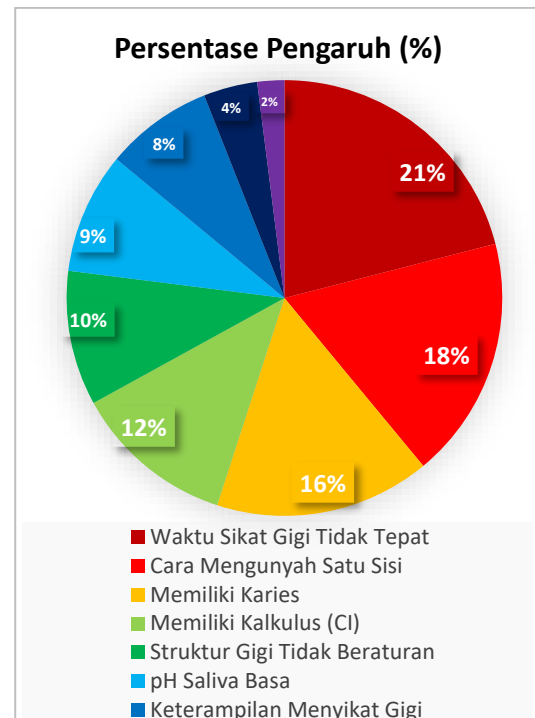


Figure 4. Influencing Factors Percentage of Calculus Risk

Based on Table 3 and Figure 2, the system successfully identified and quantified factors contributing to risk scores in the study population. It found that the most influential factor was inappropriate toothbrushing time, which contributed 21% to the total identified risk.

This discussion examines the significance of findings from a trial of the "Calrisk Dentistry" application, in which the developed system successfully classified 60 patients into four risk categories. The finding that the majority of participants fell into the moderate (40%) and high (31.7%) risk categories indicates that risk factors for calculus formation were significantly present in the study population. The presence of 13.3% of participants in the very high-risk category particularly underscores the urgency of clinical intervention for this group and validates the system's ability to identify priority cases.

Functionally, these results validate the system's effectiveness as an early screening tool. The application not only labels risks but also successfully quantifies and prioritizes the most predominant contributing factors. The identification of improper brushing time (21%), unilateral chewing (18%), and the presence of caries (16%) as the three highest risk factors provides highly specific and actionable insights.

The relationship between inappropriate brushing times and calculus formation is strong.

Brushing outside the recommended times, such as not brushing after breakfast, leaves food residue that allows plaque to develop throughout the day. Such as research by Sahoo et al (2022) revealed that rinse after breakfast can reduce *Streptococcus mutans* bacteria in 38% compared to those who didn't rinse after breakfast which the *S. mutans* only reduced at 29% [19]. Similarly, rinsing also effective in the decrease of dental plaque with p-value of 0.000 in which before rinsing, the dental plaque index was 1.31 and after rinsing, the plaque decreased to 0.47 [20]. the behavior of doesn't do tooth brushing before bed facilitates bacterial growth because saliva production decreases during sleep, making it easier for accumulated plaque to harden into calculus [21], [22].

The habit of chewing on one side also contributes significantly to the risk. The side of the mouth that is not actively used for chewing loses the natural cleaning mechanism of food movement and salivary flow. As a result, food debris, plaque, and bacteria accumulate more easily on the inactive side, increasing and accelerating the risk of tartar formation [23]. The presence of caries is often an indirect cause of one-sided chewing habits. Pain or discomfort from cavities (caries) on one side makes a person tend to only use the healthy side of the mouth for chewing. This habit not only worsens caries but also leads to plaque accumulation and calculus formation on the less-used side. These findings align with previous studies. Research by Lertpimonchai et al (2020), for example, also emphasized that improper toothbrushing behavior is a major cause of high calculus prevalence by two-to-five fold [24].

Furthermore, the development of clinical decision support systems such as "Calrisk Dentistry" aligns with the trend of utilizing digital technology in health education. The use of digital tools plays a crucial role in increasing public awareness of health care [25], [26], [27]. Therefore, the developed application functions not only as a screening tool but also as a modern and personalized educational intervention medium.

This research has several important implications. From a practical perspective, the "Calrisk Dentistry" application has the potential to become an efficient and low-cost mass screening tool for healthcare workers. From a technological perspective, this study demonstrates the successful implementation of a simple rule-based expert system in the context of dental health in Indonesia. This strengthens the external validity of this study's findings and demonstrates that the developed

algorithm is capable of capturing clinically significant risk factors. The developed scoring algorithm model can serve as a basis for the development of more complex systems in the future.

However, several limitations should be acknowledged. The relatively small sample size ($n = 60$) may restrict the generalizability of the results to a broader population. Additionally, the cross-sectional design provides only a single-time assessment and does not account for longitudinal changes in calculus risk. The absence of a control group also limits the ability to compare the system's effectiveness with conventional diagnostic methods. Despite these limitations, the study provides a solid foundation for the integration of digital decision support tools in preventive dentistry.

Conclusion

This study successfully developed and validated the *Calrisk Dentistry* web-based application as a clinical decision support tool for assessing calculus risk. The system effectively classified patients into four risk categories and identified key behavioral risk factors particularly brushing time and chewing habits as major contributors. These findings confirm that *Calrisk Dentistry* functions as a reliable digital tool for healthcare professionals, offering objective and personalized assessments that support early detection and prevention of dental calculus.

Future research is recommended to involve a larger and more diverse sample population to enhance the generalizability of the findings. Conducting longitudinal studies would allow for monitoring changes in calculus risk over time and evaluating the effectiveness of the system in real-world clinical settings. It is also advisable to include a control group using standard assessment methods to further validate the system's diagnostic accuracy. From a system development perspective, further research could focus on integrating machine learning models to improve prediction accuracy as user data grows. Adding a teleconsultation feature to directly connect high-risk patients with healthcare professionals could also be a valuable development direction.

References

- [1] K. Ari Niken Ayu Renaningtyas, D. Risnawati, and W. Haryani, "The Effect of Dental Health Education with Video Media on

- Knowledge and Interest Odontectomy Procedures,” *Jurnal Kesehatan Gigi*, vol. 12, no. 1, pp. 17–22, 2025, [Online]. Available: <http://ejournal.poltekkes-smg.ac.id/ojs/index.php/jkg/index>
- [2] F. N. Hugo, N. J. Kassebaum, W. Marcenes, and E. Bernabé, “Role of Dentistry in Global Health: Challenges and Research Priorities,” *J Dent Res*, vol. 100, no. 7, pp. 681–685, 2021, doi: 10.1177/0022034521992011.
 - [3] Kemenkes BKKP, “Survey Kesehatan Indonesia,” Jakarta, 2023.
 - [4] A. Łasica, P. Golec, A. Laskus, M. Zalewska, M. Gędaj, and M. Popowska, “Periodontitis: etiology, conventional treatments, and emerging bacteriophage and predatory bacteria therapies,” *Front Microbiol*, vol. 15, Sep. 2024, doi: 10.3389/fmicb.2024.1469414.
 - [5] S. K. Harrel, C. M. Cobb, L. N. Sheldon, M. P. Rethman, and J. S. Sottosanti, “Calculus as a Risk Factor for Periodontal Disease: Narrative Review on Treatment Indications When the Response to Scaling and Root Planing Is Inadequate,” *Dent J (Basel)*, vol. 10, no. 10, p. 195, Oct. 2022, doi: 10.3390/dj10100195.
 - [6] Y. Wei *et al.*, “Recent advances in the pathogenesis and prevention strategies of dental calculus,” *NPJ Biofilms Microbiomes*, vol. 10, no. 1, p. 56, Jul. 2024, doi: 10.1038/s41522-024-00529-1.
 - [7] C. M. Cobb, S. K. Harrel, D. Zhao, and P. Spencer, “Effect of EDTA Gel on Residual Subgingival Calculus and Biofilm: An In Vitro Pilot Study,” *Dent J (Basel)*, vol. 11, no. 1, p. 22, Jan. 2023, doi: 10.3390/dj11010022.
 - [8] S. K. Harrel, A. Yoshimura, and C. M. Cobb, “The Reevaluation of Subgingival Calculus: A Narrative Review,” *Dent J (Basel)*, vol. 13, no. 6, p. 257, Jun. 2025, doi: 10.3390/dj13060257.
 - [9] E. Raittio, R. Lopez, and V. Baelum, “Contesting the conventional wisdom of periodontal risk assessment,” *Community Dent Oral Epidemiol*, vol. 52, no. 4, pp. 487–498, Aug. 2024, doi: 10.1111/cdoe.12942.
 - [10] Dr. J. Bhuvaneswarri.MDS, Dr. JuliusAmaldas, and Dr. Ramya.V, “Risk Assessment in Periodontal Disease – A New Way of Predicting the Periodontal Disease Outcome,” *Int J Life Sci Pharma Res*, Sep. 2023, doi: 10.22376/ijlpr.2023.13.5.L228-L234.
 - [11] G. V. de O. Fernandes, G. Mosley, A. C. Cañizares, J. C. H. Fernandes, and R. Muller, “Standardizing clinical evaluations of periodontal condition to guide assessments and diagnoses using the Periodontal Assessment Protocol (GF-PAPro),” *Advances in Clinical and Experimental Medicine*, vol. 34, no. 7, pp. 1069–1077, Jul. 2025, doi: 10.17219/acem/207502.
 - [12] L. J. Maciel, L. D. Boyd, L. J. Giblin-Scanlon, and J. Vineyard, “Utilization of Periodontal Risk Assessment Tools in the Clinical Setting: Knowledge, attitudes and practice behaviors of dental hygienists,” *The Journal of Dental Hygiene*, vol. 94, no. 6, pp. 16–24, 2020.
 - [13] L. Dwi Apriyanti, D. Fatmasari, T. Wiyatini, and M. Djamil, “Interactive Media with Android Based Sign Language as an Effort to Improve Dental Health Maintenance Behaviour and Decrease Debris Index in Deaf Children,” *Jurnal Kesehatan Gigi*, vol. 12, no. 1, pp. 1–7, 2025, [Online]. Available: <http://ejournal.poltekkes-smg.ac.id/ojs/index.php/jkg/index>
 - [14] O. Kalmus, K. Smits, M. Seitz, C. Haux, B.-P. Robra, and S. Listl, “Evaluation of a Digital Decision Support System to Integrate Type 2 Diabetes Mellitus and Periodontitis Care: Case-Vignette Study in Simulated Environments,” *J Med Internet Res*, vol. 25, p. e46381, Oct. 2023, doi: 10.2196/46381.
 - [15] M. T. G. Thanh, N. Van Toan, V. T. N. Ngoc, N. T. Tra, C. N. Giap, and D. M. Nguyen, “Deep Learning Application in Dental Caries Detection Using Intraoral Photos Taken by Smartphones,” *Applied Sciences*, vol. 12, no. 11, p. 5504, May 2022, doi: 10.3390/app12115504.
 - [16] T.-J. Lin *et al.*, “Auxiliary Diagnosis of Dental Calculus Based on Deep Learning and Image Enhancement by Bitewing Radiographs,” *Bioengineering*, vol. 11, no. 7, p. 675, Jul. 2024, doi: 10.3390/bioengineering11070675.
 - [17] B. Khalili and A. W. Smyth, “SOD-YOLOv8—Enhancing YOLOv8 for Small Object Detection in Aerial Imagery and Traffic Scenes,” *Sensors*, vol. 24, no. 19, p. 6209, Sep. 2024, doi: 10.3390/s24196209.
 - [18] P. Lucchi *et al.*, “Introducing the Index of Caries Risk (ICR): A Comparative Study on a Novel Tool for Caries Risk Assessment in Pediatric Patients,” *Children*, vol. 11, no. 10, p. 1166, Sep. 2024, doi: 10.3390/children11101166.

- [19] S. R. Sahoo, D. B. Nandini, P. S. Basandi, M. Selvamani, and M. Donoghue, "A Comparison of Pre- and Postbreakfast Tooth Brushing in Caries Prevention through the Estimation of Streptococcus mutans Counts," *J Microsc Ultrastruct*, vol. 10, no. 4, pp. 168–173, Oct. 2022, doi: 10.4103/jmau.jmau_90_21.
- [20] S. S. Habibah, D. Danan, and R. Ulfah, "The Effect Of Young Coconut Water Mouth (Cocus Nucifera L) Against Dental Plaque," *Jurnal Kesehatan Gigi*, vol. 10, no. 2, pp. 97–101, Dec. 2023, doi: 10.31983/jkg.v10i2.10356.
- [21] G. Kitsaras, M. Goodwin, M. P. Kelly, and I. A. Pretty, "Bedtime Oral Hygiene Behaviours, Dietary Habits and Children's Dental Health," *Children*, vol. 8, no. 5, p. 416, May 2021, doi: 10.3390/children8050416.
- [22] M. Funahara, R. Yamaguchi, H. Honda, M. Matsuo, W. Fujii, and A. Nakamichi, "Factors affecting the number of bacteria in saliva and oral care methods for the recovery of bacteria in contaminated saliva after brushing: a randomized controlled trial," *BMC Oral Health*, vol. 23, no. 1, p. 917, Nov. 2023, doi: 10.1186/s12903-023-03676-7.
- [23] T. S. Abbas, G. M. Al-hamdani, and K. A. Mahdi, "Impact of Unilateral Chewing Habit on Temporomandibular Joint Disorders: A Comparative Study," *Journal of Indian Academy of Oral Medicine and Radiology*, vol. 37, no. 3, pp. 328–331, Jul. 2025, doi: 10.4103/jiaomr.jiaomr_160_25.
- [24] A. Lertpimonchai, S. Rattanasiri, S. Arj-Ong Vallibhakara, J. Attia, and A. Thakkinstian, "The association between oral hygiene and periodontitis: a systematic review and meta-analysis," *Int Dent J*, vol. 67, no. 6, pp. 332–343, Dec. 2017, doi: 10.1111/idj.12317.
- [25] J. C. Ferreira, L. B. Elvas, R. Correia, and M. Mascarenhas, "Empowering Health Professionals with Digital Skills to Improve Patient Care and Daily Workflows," *Healthcare*, vol. 13, no. 3, p. 329, Feb. 2025, doi: 10.3390/healthcare13030329.
- [26] M. Senbekov *et al.*, "The Recent Progress and Applications of Digital Technologies in Healthcare: A Review," *Int J Telemed Appl*, vol. 2020, pp. 1–18, Dec. 2020, doi: 10.1155/2020/8830200.
- [27] Sadhana Kandavel, Anita M, Vidhya Rehka U., and Sathyasri P, "Mobile Smartphone Apps Fororal Dental Health-Review Article,"

European Journal of Molecular & Clinical Medicine, vol. 07, no. 2, p. 6755, 2020, doi: 10.2196/11432.